



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE VENKATESHWARA FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

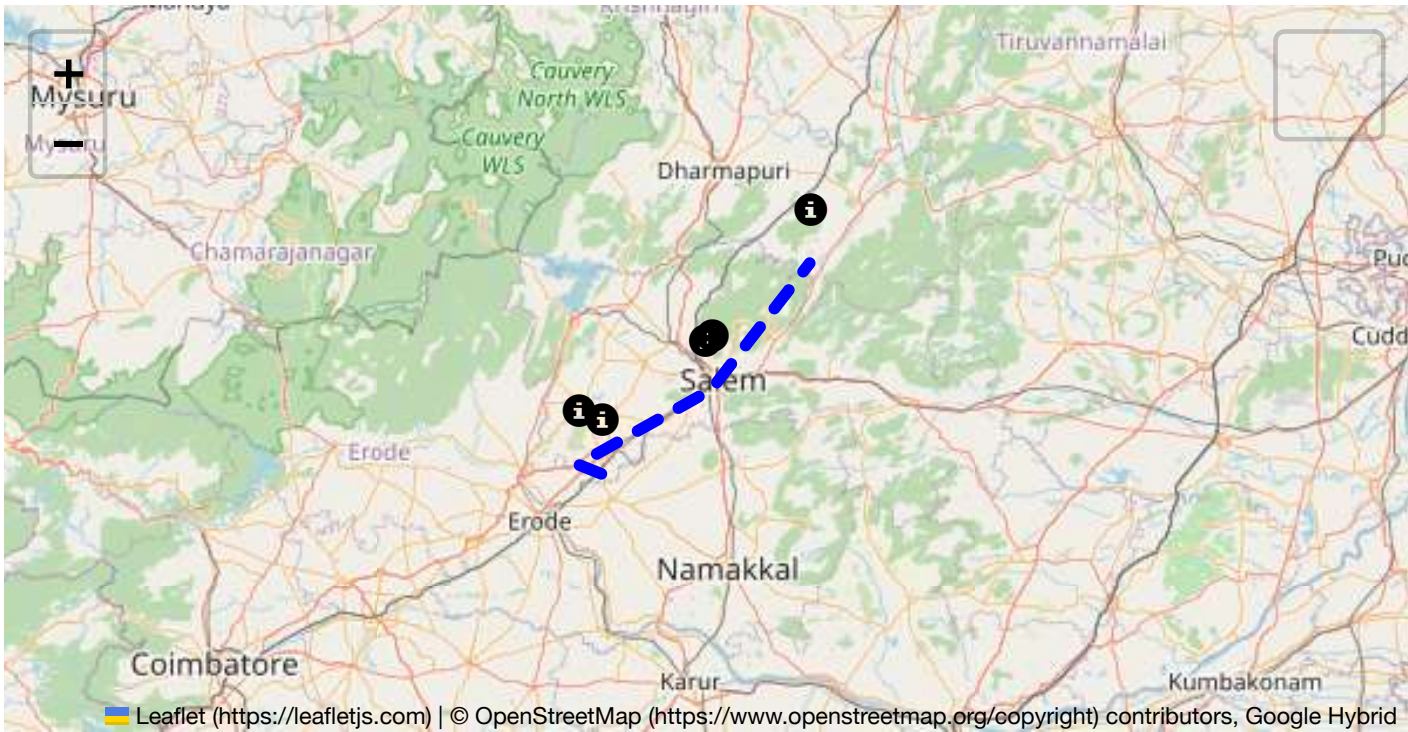
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

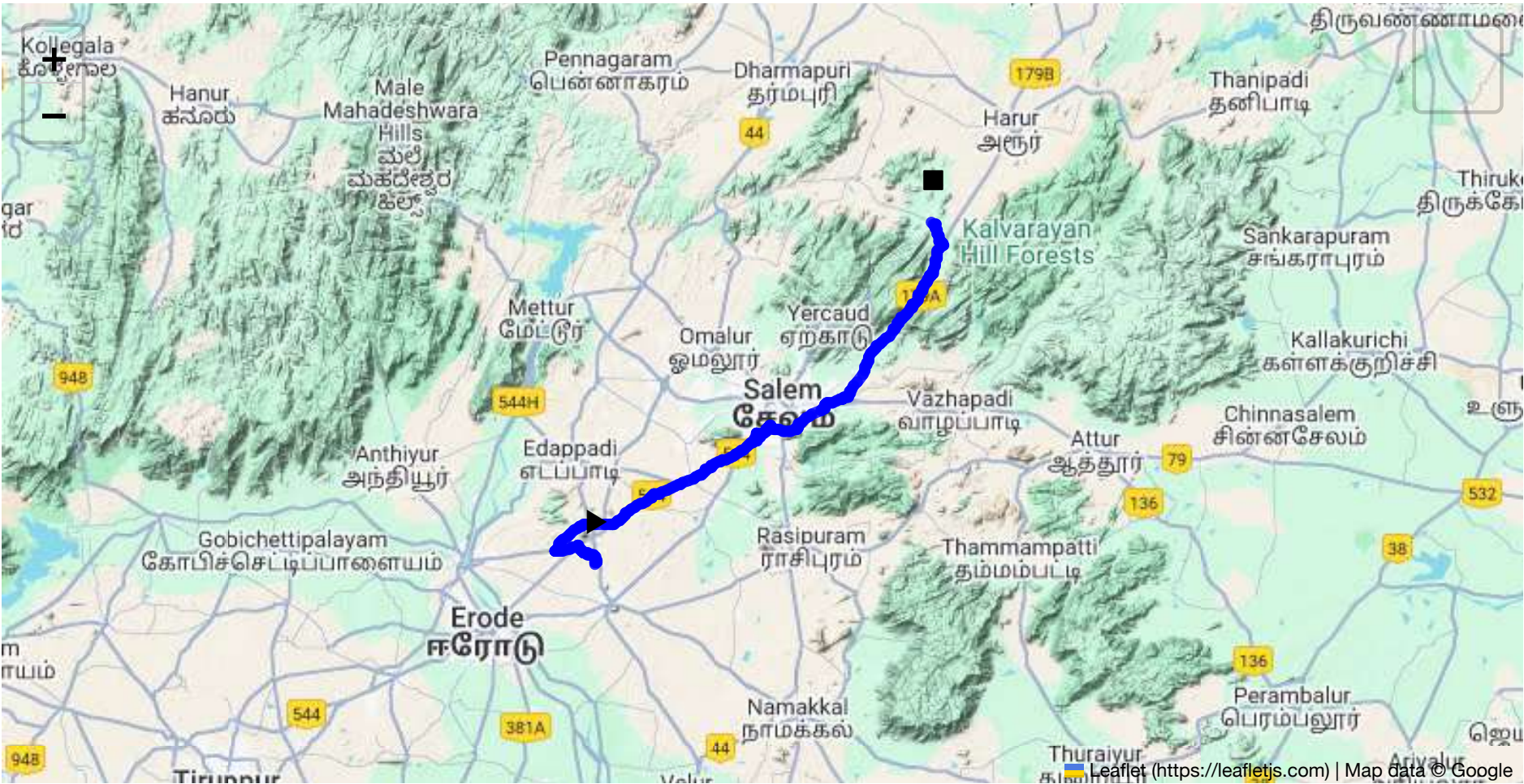


Route Summary:
Total Distance: 99.64 km
Estimated Duration: 2.5 hours

Adjusted Duration (Heavy Vehicle): 3.1 hours

Start: (11.4381, 77.8734)

End: (11.920037, 78.359218)



Welcome to the Journey Risk Management Study

- Overview of the Route Map:** The route runs from Sangagiri to Kolimekkanur, passing through significant checkpoints such as Bhavani Main Road in Padaiveedu and Sankari Main Road in Salem. This route is primarily on regional roads with moderate infrastructure, connecting agricultural and industrial zones in Tamil Nadu.
- Typical Weather Conditions and Potential Weather-Related Hazards:** The region experiences a tropical climate, with hot summers, a monsoon season from June to September, and mild winters. During monsoon, heavy rainfall may cause flooding or landslides, leading to road blockages and deteriorated road conditions, especially on less maintained rural roads.
- Analysis of Traffic Patterns:** Traffic congestion is typically higher during morning and evening rush hours (8:00-10:00 AM and 5:00-7:00 PM). Salem is a major transit point, and congestion can occur especially near industry zones and marketplaces, which could delay travel times.
- Assessment of Road Quality and Infrastructure:** The road quality varies, with main roads being in relatively good condition but secondary roads showing signs of wear and varying maintenance levels. Expect potholes or uneven pavement, especially outside major towns.
- Suggestions for Alternative Routes for Emergencies:** In case of obstructions, rerouting through NH 44 or NH 544 from Salem can provide alternative pathways. These national highways are better maintained and equipped to support detours but may increase overall travel time.
- Summary of Local Regulations Affecting Hazardous Material Transport:** Strict adherence to regulations regarding travel times for vehicles carrying hazardous materials is required, often limited to specific hours to avoid peak traffic times. Proper documentation and safety measures must be ensured to comply with state-enforced transport laws.
- Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** The area has had occasional incidents related to heavy vehicle collisions, generally attributed to poor weather and road

conditions, and driver fatigue. While incidents involving hazardous materials are rare, vigilance is recommended.

8. **Environmental Considerations and Sensitive Areas:** The route passes through rural farmland and sensitive ecological zones, where ecosystems can be affected by hazardous spills. Drivers should be cautious near water bodies and designated ecological areas to prevent contamination.
9. **Analysis of Communication Coverage:** Communication coverage is generally adequate in urban and suburban stretches; however, dead zones are possible in rural segments, affecting mobile network reliability. Carrying communication equipment with satellite backup is advisable.
10. **Estimated Emergency Response Times for Different Route Segments:** Response times in urban areas like Salem are relatively fast, approximately within 15-30 minutes due to the presence of medical and emergency facilities. In rural stretches, response times may be delayed, ranging from 45 minutes to over an hour.
11. **Overall Summary of Risk Assessment:** The route presents a moderate level of risk for transporting hazardous materials. Key concerns are weather-related impacts, road quality, and varying traffic congestion. Preparedness through route planning, awareness of emergency contacts, and proactive compliance with transport regulations are crucial to reduce potential hazards. Regular updates on weather conditions and road status are recommended for safe transit along this path.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Roundabout	High	11.63247, 78.12734	15 KM/Hr
2	U-Turn	High	11.6193714, 78.1124012	10 KM/Hr
3	U-Turn	High	11.618013, 78.1109532	10 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	Medium	11.44900, 77.87406	30 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45853, 77.85139	30 KM/Hr
9	Turn	Medium	11.45906, 77.85132	30 KM/Hr
10	Turn	High	11.46307, 77.84941	15 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45636, 77.81678	15 KM/Hr
13	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
14	Turn	High	11.62130, 78.10776	15 KM/Hr
15	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
16	Turn	Medium	11.61224, 78.10469	30 KM/Hr
17	Blind Spot	Blind Spot	11.61928, 78.11242	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Blind Spot	Blind Spot	11.61801, 78.11095	10 KM/Hr
19	Turn	Medium	11.61817, 78.11102	30 KM/Hr
20	Turn	High	11.63300, 78.12746	15 KM/Hr
21	Turn	High	11.63306, 78.12731	15 KM/Hr
22	Turn	Medium	11.66372, 78.20649	30 KM/Hr
23	Turn	Medium	11.67130, 78.22990	30 KM/Hr
24	Blind Spot	Blind Spot	11.67111, 78.23134	10 KM/Hr
25	Turn	High	11.67203, 78.23152	15 KM/Hr
26	Turn	Medium	11.67445, 78.23920	30 KM/Hr
27	Turn	Medium	11.80448, 78.33329	30 KM/Hr
28	Turn	Medium	11.81243, 78.34087	30 KM/Hr
29	Turn	Medium	11.83483, 78.35295	30 KM/Hr
30	Turn	Medium	11.88901, 78.37646	30 KM/Hr
31	Turn	Medium	11.88930, 78.37653	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
17	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
20	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
21	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
22	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
23	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
24	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
25	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
26	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium
31	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
33	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
34	hospital	Pappireddipatti Government Hospital	11.9179195, 78.3646393	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
8	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium
28	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
29	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
32	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.63247, 78.12734



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.6193714, 78.1124012



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.618013, 78.1109532



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



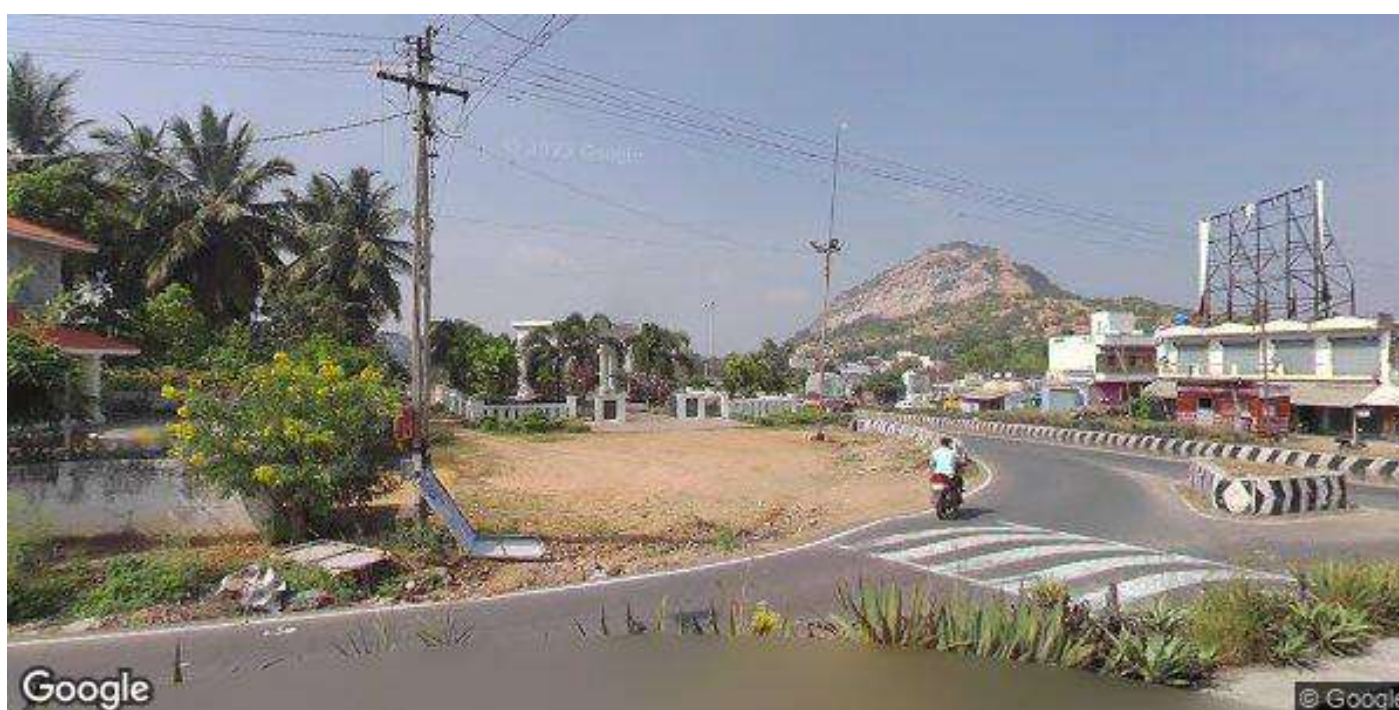
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61224, 78.10469



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61928, 78.11242



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61801, 78.11095



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61817, 78.11102



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63306, 78.12731



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66372, 78.20649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67130, 78.22990



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67111, 78.23134



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67203, 78.23152



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67445, 78.23920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80448, 78.33329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81243, 78.34087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83483, 78.35295



Risk Type: Turn
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Speed Limit: 30 KM/Hr
Coordinates: 11.88901, 78.37646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88930, 78.37653

